Algebraic Number Theory NMAL431
Exercise Sheet 2

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1. Let $f(x) = x^3 - 3x + 4$.
   (a) Show that there exists a unique $x \in \mathbb{Z}_7$ such that $f(x) = 0$.
   (b) Show that $f(x) = 0$ has no solutions in $\mathbb{Z}_3$ or $\mathbb{Z}_5$.
   (c) Show that there exists a unique $x \in \mathbb{Z}_2$ such that $f(x) = 0$.

2. Show that $(x^2-2)(x^2-17)(x^2-34)$ has a root in $\mathbb{R}$ and in $\mathbb{Q}_p$ for every prime p, but no root in $\mathbb{Q}$.

3. Let p be a prime, and let $n \geq 1$ be coprime to p. Show that the homomorphism $1+p\mathbb{Z}_p \to 1+p\mathbb{Z}_p$ given by $x \mapsto x^n$ is a bijection. [*Hint*: Use Hensel's lemma]

4. (a) Let p be an odd prime and let $\alpha \in \mathbb{Z}_p^\times$. Show that α lies in $(\mathbb{Z}_p^\times)^p$ if and only if $\alpha \bmod p^2\mathbb{Z}_p \in ((\mathbb{Z}/p^2\mathbb{Z})^\times)^p$.
   (b) Let $p = 2$ and let $\alpha \in \mathbb{Z}_2^\times$. Show that α lies in $(\mathbb{Z}_p^\times)^p$ if and only if $\alpha \bmod p^3\mathbb{Z}_p \in ((\mathbb{Z}/p^3\mathbb{Z})^\times)^p$.

   [*Hint:* Use Hensel's lemma]

5. Let $M \geq 2$. Prove that there exists an isomorphism

$$\varprojlim_i \mathbb{Z}/M^i\mathbb{Z} \cong \prod_{p|M} \mathbb{Z}_p$$

   [*Hint:* Use the Chinese remainder theorem]

6. Let p be a prime.
   (a) Prove that there exists a continuous surjection $\mathbb{Z}_p \to [0,1]$.
   (b) Show that there does *not* exist a continuous surjection $[0,1] \to \mathbb{Z}_p$.

7. Let p be a prime, let $x \in \mathbb{Q}_p$, and write $x = \sum_{n \geq N} a_n p^n$ with $a_n \in \{0, 1, \ldots, p-1\}$ for each $n \geq N$. Show that $x \in \mathbb{Q}$ if and only if the sequence (a_n) is eventually periodic.

8. Show that $\mathbb{Q}_2^\times/(\mathbb{Q}_2^\times)^2$ has order 8, with representatives $\pm 1, \pm 2, \pm 5, \pm 10$. Deduce that $\mathbb{Q}_2$ has exactly 7 quadratic extensions up to isomorphism, and write each one down.